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SilicaFlux ASIC Production Process: From SiO_2 to Packaged Silicon

Executive summary

The technically sensible route for SilicaFlux is **not** to fabricate a semiconductor directly from silica. Silicon dioxide, SiO_2, is an insulator; the manufacturing chain begins by reducing high-purity quartz to metallurgical-grade elemental silicon, chemically refining that material to electronic-grade polysilicon, growing a single-crystal silicon ingot, slicing and polishing wafers, and only then fabricating CMOS transistors and interconnects on those wafers. Industrial silicon production uses carbothermic reduction of quartz in submerged electric-arc furnaces, while semiconductor-grade polysilicon is normally produced through chlorosilane chemistry and the Siemens process. WACKER reports semiconductor polysilicon purity at approximately 99.9999999999%, with critical impurities controlled into the parts-per-trillion regime. [1]

For **the current SilicaFlux architecture**, the most important manufacturing fact is its memory footprint. Direct inspection of `SilicaFlux2.0/SilicaFlux2.5/silicaflux_top.sf` shows default parameters of two core complexes × four cores, eight LLC banks, 2 MiB of L2 per complex, and **4 MiB per LLC bank**. That implies approximately **36 MiB of L2+LLC SRAM before counting scratchpads, tags, TLB structures, register files or other memories**. The top-level also exposes separate core, memory, host, RTC and JTAG clock domains and has explicit DFT inputs. The project's earlier assessment describes the broader repository as roughly 30,768 lines across 179 files and notes that verification is substantially less mature than an industrial production ASIC flow. filecite turn0file0 L1-L18

That memory footprint changes the process-node decision dramatically. A full default SilicaFlux implementation is poorly matched to 180 nm and unattractive at 65 nm because embedded SRAM would dominate die area, capacitance and cost. **A mature 28 nm CMOS process is the best default recommendation** if the aim is to fabricate the complete architecture rather than a drastically reduced demonstrator. GlobalFoundries, for example, continues to manufacture 28 nm platforms on 300 mm wafers, and its 28 nm offerings include production-proven low-power/HKMG technologies and broad IP ecosystems. [2] A 22 nm FD-SOI variant would also merit consideration for an aggressively power-constrained version because body biasing and very-low-leakage SRAM are available, but it is not necessary merely to make SilicaFlux viable. [3]

Seven nanometres would provide much higher density and potentially lower dynamic energy per operation, but it is economically unjustified unless SilicaFlux is redesigned for substantially higher frequency, much larger vector/tensor throughput, or sufficiently high production volume to amortise expensive masks, IP, EDA and verification. TSMC states that its N7 technology offers up to roughly 3× logic density, 30% speed gain and 55% power reduction relative to N16 under its comparison conditions, but those benefits come with dramatically more demanding physical design and manufacturing. [4]

My recommended production strategy is therefore:

Decision	Recommended starting point
Wafer material	Electronic-grade CZ silicon, 300 mm for 28 nm
CMOS node	**28 nm mature planar/HKMG CMOS**
First prototype	MPW shuttle before dedicated masks
Production configuration	Reduce the default cache capacity for first silicon unless a ≥50–100 mm² die is commercially acceptable
SRAM	Foundry-compiled SRAM macros, never synthesised flip-flop arrays
Package	Flip-chip BGA for the full device; wire-bond package only for a reduced low-I/O prototype
Clocking	One PLL near the package/I/O edge feeding hierarchical clock domains; CTS performed after placement
CPU complexes	Central-left/right symmetric islands
Tensor/vector engines	Adjacent to the relevant L2/scratchpad banks
LLC	Distributed SRAM banks around the compute fabric rather than one monolithic block
Memory interfaces	Along the die edge nearest the package balls assigned to external memory
Power	Multiple VDD/VSS bumps, perimeter rings plus dense internal mesh
Test	Scan, ATPG, MBIST/BISR for every SRAM family, JTAG boundary scan and on-chip process monitors
Production gate	No tapeout until RTL verification, STA, CDC/RDC, DRC, LVS, IR-drop, EM, DFM and test coverage have closed

The raw-material chain and the ASIC-design chain are largely separate industrial disciplines. **It would make no economic sense for a SilicaFlux project to construct its own quartz-to-wafer or wafer-fabrication facility.** For perspective, TSMC's 22/28 nm Japanese fab was announced at approximately US$7 billion of initial capital expenditure, while its two Arizona advanced-node fabs were announced at roughly US$40 billion combined. [5] SilicaFlux should operate as a **fabless ASIC project**, purchasing qualified wafers and foundry manufacturing services.

Materials chain from quartz to transistor-grade wafers

Reduction of SiO_2

The starting material should be high-purity lump quartz or quartzite rather than arbitrary sand. Silicon producers use quartz plus carbonaceous reductants in a submerged electric-arc furnace. The simplified net reaction is:

$$\mathrm{SiO_2 + 2C \rightarrow Si + 2CO}$$

The real furnace chemistry proceeds through SiO and SiC intermediates and operates at temperatures approaching roughly 2,000 °C. Industrial silicon production typically consumes about 10–14 MWh of electricity per tonne of silicon. [6]

The product of this step is metallurgical silicon, typically around 99% purity rather than semiconductor purity. Iron, aluminium, calcium and titanium are representative contaminants. [7] This distinction is critical: attempting to grow an ASIC wafer from metallurgical silicon would produce unacceptable electrical variability, minority-carrier degradation and oxide reliability problems.

Conversion to electronic-grade polysilicon

The preferable industrial route is chemical purification, not simply repeated melting. Metallurgical silicon is reacted into a volatile chlorosilane—principally trichlorosilane, $SiHCl_3$—which can be purified by repeated distillation. The purified precursor and hydrogen are then introduced into a Siemens reactor containing heated silicon seed rods at approximately 1,000 °C, where high-purity silicon deposits onto the rods. [8]

This chemical route is powerful because impurities can be separated through volatility differences before silicon is re-deposited. WACKER describes its semiconductor-grade polysilicon as 99.9999999999% pure, with impurity concentrations in the ppt regime; semiconductor material is additionally surface-etched and packaged under clean conditions after crushing because crushing itself can contaminate the surface. [9]

For SilicaFlux, the important contaminant categories are:

- electrically active dopants such as boron and phosphorus, which alter wafer resistivity;
- transition and heavy metals, which introduce recombination centres and leakage;
- oxygen and carbon, whose concentrations affect defects and precipitation;
- particles and surface metals, which become catastrophic as feature dimensions shrink.

SUMCO describes polysilicon used for CZ growth as having metallic impurities at no more than a few parts per billion before deliberate dopants are introduced. [10]

Zone refining and float-zone material

Zone refining exploits segregation coefficients: a narrow molten region is moved through a silicon rod so impurities preferentially remain in the liquid phase and migrate towards one end. It is capable of exceptional purification, but **it is not the normal material path I would choose for a mainstream 28 nm digital SoC**.

Float-zone crystal growth similarly avoids contact between molten silicon and a quartz crucible. Siltronic notes that FZ material therefore contains less oxygen and can provide very high resistivity and long carrier lifetime; this makes it particularly valuable for power and high-voltage semiconductor applications. [11] Those properties bring little benefit to an ordinary dense CMOS CPU/vector/tensor SoC and FZ wafer diameter/availability are less favourable.

For SilicaFlux, **CZ or magnetic-CZ is the appropriate baseline**. It is the mainstream path to large-diameter logic wafers and allows tightly controlled doping and resistivity.

Czochralski single-crystal growth

Electronic-grade polysilicon is loaded into an ultrapure quartz crucible and melted at roughly 1,420 °C. Controlled quantities of boron, phosphorus or another dopant establish the target conductivity and resistivity. A single-crystal seed is dipped into the melt and slowly pulled while seed and crucible rotation control diameter and crystal quality. [12]

For a 28 nm SilicaFlux product, a typical procurement specification would be a **300 mm CZ prime wafer** satisfying the foundry's own crystal-defect, resistivity, oxygen, flatness, edge and contamination limits. These limits should come from the selected foundry rather than being invented by the chip designer.

The ingot subsequently undergoes diameter grinding, wafer slicing, edge rounding, lapping or grinding, chemical etching, cleaning and chemical-mechanical polishing. Epitaxial deposition may then be performed where required by the foundry process. Siltronic's current process description includes CZ/FZ growth, ingot grinding, multi-wire slicing, edge rounding, identification, lapping, cleaning/etching, polishing and optional epitaxy. [13]

The division of responsibility should therefore be:

```
SilicaFlux company
        │
        ├── owns RTL, verification, physical design and product specification
        │
Wafer supplier
        ├── quartz-derived electronic-grade polysilicon
        ├── crystal growth
        └── polished prime wafers
        │
Foundry
        ├── CMOS FEOL
        ├── MOL
        ├── BEOL
        ├── wafer probe support
        └── diced wafer / known-good die
        │
OSAT
        ├── bump / assembly
        ├── package
        └── final test
```

Trying to vertically integrate the first two industrial stages for a single ASIC would be financially irrational.

Process-node selection for SilicaFlux

Architectural implications of the RTL

The uploaded code is not merely a small educational RISC-V controller. The newest top-level inspected in the supplied archive exposes:

- `NUM_CPLX = 2`;
- `CORES_PER_CPLX = 4`;
- `NUM_LLC_BANKS = 8`;
- `L2_SIZE_KB = 2048`;
- `LLC_BANK_SIZE_KB = 4096`;
- vector and tensor units enabled;
- ECC, trace, PMU and debug enabled;
- two external memory ports and two DMA ports;
- multiple clock domains.

Consequently, the default system contains **8 cores, 4 MiB of aggregate private/complex L2 and 32 MiB of LLC**. The earlier project assessment also reports a substantial modular RTL base but identifies incomplete verification infrastructure as the principal industrialisation gap. filecite turn0file0 L147-L176

The tensor engine itself is physically significant. `tensor_unit.sf` defines a default 4×4 matrix multiply with 16 output-stationary lanes, radix-4 Booth multiplication, carry-save compression and a 20-stage pipeline capable, according to its RTL comments, of accepting one 4×4 operation each cycle once filled. This should be treated as a regular datapath region during placement rather than allowed to scatter throughout the standard-cell sea.

The default `compute_tile.sf` additionally contains scalar and vector scratchpad regions, two scalar ALUs and a tensor engine. Those structures suggest **locality-oriented floorplanning**, with execution blocks physically adjacent to their scratchpads and register structures.

Node comparison

The following table is the decision model I would use. The area and production-wafer figures are **engineering planning ranges, not foundry quotations**; actual figures depend on SRAM compiler options, metal stack, threshold-voltage mix, design density, masks, wafer volume and commercial agreements. Public prototype prices demonstrate the relative escalation but do not directly equal volume wafer prices. EUROPRACTICE's 2026 public pricing, for example, lists UMC 180 nm MPW blocks at €16,750, UMC 65 nm blocks at €43,750, and GF 28 nm SLPe at €12,430/mm² under its applicable MPW terms. [14]

Attribute	180 nm	65 nm	**28 nm**	7 nm
Transistor style	Planar bulk CMOS	Planar CMOS	Planar/HKMG or FD-SOI	FinFET
Typical wafer diameter	200/300 mm depending process	Usually 300 mm	300 mm	300 mm

Attribute	180 nm	65 nm	**28 nm**	7 nm
Relative logic density	Very low	Moderate	High	Very high
36+ MiB embedded SRAM suitability	**Poor**	Possible but large	**Good**	Excellent
Leakage-management difficulty	Low	Moderate	Moderate-high	High
Physical-design difficulty	Low	Moderate	High	Very high
Mask/NRE burden	Lowest	Moderate	High	Extremely high
DFM sensitivity	Low-moderate	Moderate	High	Very high
Package likely needed for full SilicaFlux	Large BGA	BGA	Flip-chip BGA	Flip-chip BGA
Indicative production-wafer planning allowance*	~$0.8k–2k	~$2k–4k	~$3k–7k	~$10k–20k+
Best SilicaFlux use	Tiny reduced demo	Reduced/ product prototype	**Full implementation**	High-volume/ high-performance derivative

*These wafer ranges are deliberately coarse budget assumptions inferred from node complexity and public shuttle economics; commercial wafer prices are normally contractual and volume-dependent. The public sources should be used as boundary evidence rather than treated as a foundry quotation. [14]

GlobalFoundries explicitly positions mature CMOS from ≥180 nm through 22 nm for cost-sensitive edge products, while its 28 nm process family supports integrated digital, analogue and embedded-memory applications. [15] TSMC describes N7 as an HPC-class process with major power, performance and density gains over 16 nm. [16]

Why 180 nm is wrong for the default chip

A reduced SilicaFlux demonstrator—with perhaps one CPU, a few tens or hundreds of KiB of SRAM and one tensor/vector block—could be manufactured economically at 180 nm. The **default full cache hierarchy cannot sensibly be carried across unchanged**.

Even ignoring logic, 36 MiB represents about 302 million stored bits. An SRAM macro requires at least six transistors per bit in conventional 6T form, so the storage arrays alone correspond to roughly 1.8 billion SRAM transistors before decoders, sense amplifiers, redundancy, ECC, tag RAMs, logic and routing. That is precisely the workload for which advanced embedded-memory density matters.

Why 65 nm is credible but not optimum

At 65 nm, SilicaFlux becomes physically plausible, and the node has mature libraries, relatively forgiving routing and far lower NRE than leading-edge FinFET. It would be an excellent node for a **4–8 MiB cache version** or a research production ASIC where frequency targets are moderate.

For the current 36 MiB default, however, die area and SRAM power are likely to dominate. A 65 nm implementation could therefore become less economical per functional die than a denser 28 nm solution even if wafer processing itself is cheaper.

Why 28 nm is the baseline recommendation

Twenty-eight nanometres is old enough to be highly mature but dense enough to make the SilicaFlux cache hierarchy realistic. GF reports production 28 nm on 300 mm lines, while its 28SLP/28SLPe technologies were designed for low-power/high-integration SoCs and include mature design enablement. [17]

It also avoids several costs of 7 nm:

- FinFET-specific physical-design complexity;
- substantially more expensive mask sets;
- greater dependence on licensed high-speed PHY and sign-off IP;
- tighter variation, EM and signal-integrity margins;
- EUV/multiple-patterning-related process complexity.

This makes 28 nm a much better **risk-adjusted first commercial node**.

When 7 nm becomes justified

Seven nanometres should be reconsidered if synthesis and post-layout profiling establish that SilicaFlux requires, for example, multi-GHz clocking, far more compute tiles or tens of TOPS at a strict power envelope. TSMC states that N7 provides up to 3× N16 logic density and significant speed/power advantages. [4]

But shrinking the RTL without first industrialising its verification environment would be backwards: advanced-node respins are the most expensive place to discover an architectural or CDC bug.

RTL-to-GDS architecture and SilicaFlux floorplan

Pre-tapeout transformation

The manufacturing process cannot begin from SystemVerilog alone. SilicaFlux must first be converted through:

```
RTL
 │
 ├─ lint / elaboration
 ├─ simulation + formal
 ├─ CDC / RDC
 ├─ synthesis
```

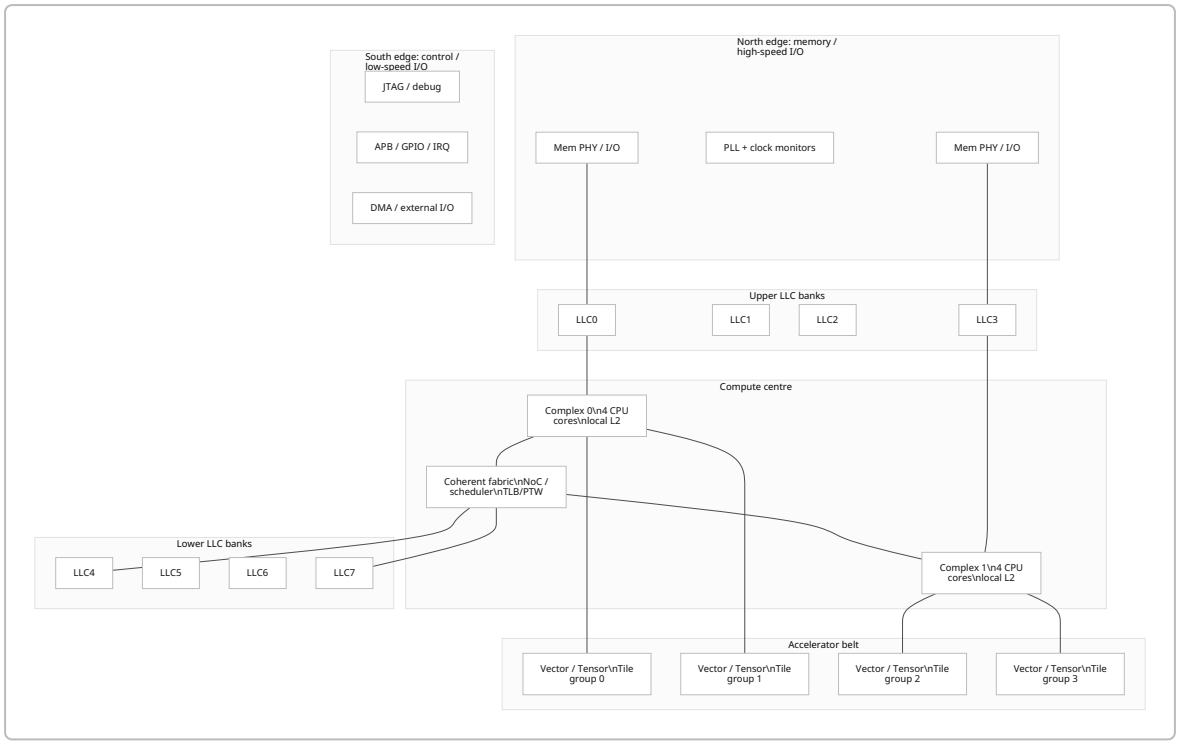
```
 ▼
technology-mapped netlist
 │
 ├─ DFT insertion
 ├─ MBIST/BISR
 ├─ floorplanning
 ├─ power planning
 ├─ placement
 ├─ CTS
 ├─ route
 ├─ parasitic extraction
 ▼
post-route database
 │
 ├─ STA / SI
 ├─ IR-drop / EM
 ├─ DRC
 ├─ LVS
 ├─ antenna / density / DFM
 └─ final ECO
 ▼
GDSII/OASIS + mask data
```

Static timing analysis evaluates all timing paths rather than merely the stimulus exercised in simulation; production closure therefore requires setup/hold, recovery/removal and clock-related paths to pass across the selected process/voltage/temperature corners. [18] DRC checks that the physical layout satisfies the selected foundry's manufacturability rules, while LVS compares the extracted physical connectivity and devices with the intended circuit/netlist. [19]

The open SkyWater and GF PDK documentation illustrates why these checks are process-specific: PDKs contain geometric rules, layer definitions, SRAM-specific rules and physical-verification decks rather than one universal set of CMOS dimensions. [20]

Recommended SilicaFlux die organisation

The floorplan should exploit locality rather than simply spreading blocks according to area.



This is a conceptual floorplan, not a dimensioned GDS layout.

The **eight LLC banks should be physically distributed**, preferably four above and four below the coherent fabric. The aim is to reduce average wire length and avoid routing all cache traffic to one edge. The coherence crossbar/snoop filter should occupy the centre so that no core complex has a systematic topology advantage.

Each CPU complex should sit immediately beside its 2 MiB L2. The vector and tensor execution structures should sit on the L2/scratchpad-facing side of each compute region. The `tensor_unit` contains wide multiplier/reduction trees, while `compute_tile` exposes local scalar/vector scratchpad regions; separating these physical blocks would increase data-bus capacitance and congestion.

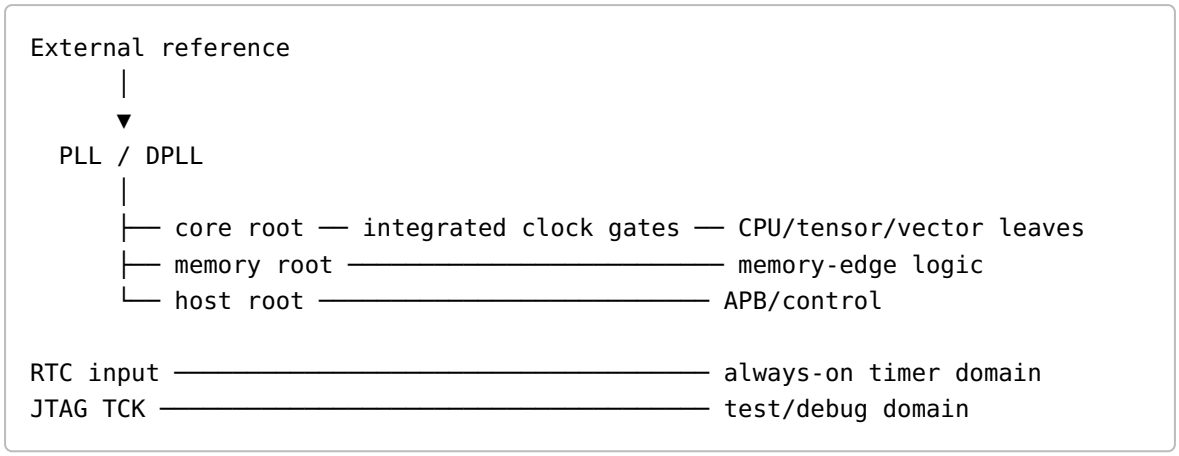
The **PLL should not sit in the noisy centre of the tensor datapath**. Place it close to a clean analogue/power region near the perimeter, surrounded by appropriate guard rings and well isolation according to the process design manual. Its clock output can then feed central clock-generation points.

Cadence's implementation documentation describes clock-tree synthesis as a joint physical optimisation problem balancing skew, insertion delay, power and routing obstacles, and modern flows explicitly integrate floorplanning, power planning, CTS, routing, parasitic extraction, IR-drop and sign-off. [21]

Clock architecture

The RTL already distinguishes `clk_core_i`, `clk_mem_i`, `clk_host_i`, `clk_rtc_i` and `jtag_tck_i`. That should be preserved physically rather than flattened into one unmanaged tree.

Recommended implementation:



Clock gating should use foundry/library-qualified integrated clock-gating cells. Cross-domain signals should use explicitly verified CDC structures, FIFOs or handshakes. The RTL comments indicate dedicated synchroniser/CDC concepts, but these must be confirmed with structural CDC analysis before tapeout rather than trusted solely from source intent.

Power distribution

A chip containing eight cores, SRAM and multiply-heavy tensor logic will have significant local transient current. The die should therefore use:

- peripheral VDD/VSS rings;
- a dense top-metal mesh;
- regular via arrays between power levels;
- separate or filtered PLL/analogue rails;
- distributed package bumps rather than edge-only power pins for the production configuration;
- local decoupling capacitance around tensor/vector blocks;
- SRAM power straps sized from compiler current models;
- power-gated islands only if the isolation, retention and sequencing logic are fully verified.

IR-drop and electromigration closure must use extracted post-route current profiles, including worst-case tensor/vector activity. Cadence includes IR-drop and power-distribution analysis as explicit sign-off stages in its digital implementation curriculum and tool flow. [22]

DFT and test structures

SilicaFlux already has `test_mode_i` and `scan_en_i` at its advanced top level, which is a useful architectural starting point but **not evidence of completed DFT**.

The production implementation should add:

1. full or very high scan coverage of flip-flops;
2. multiple scan chains to constrain tester time;
3. stuck-at and transition ATPG;
4. memory BIST for every L2, LLC, scratchpad, TLB and ROM/RAM macro;
5. SRAM redundancy/BISR if supported by the compiler;
6. JTAG/boundary scan;
7. clock-controller test modes;
8. PLL bypass/test access;

9. process-monitor structures, ring oscillators and representative device chains;
10. dedicated scribe-line process control monitors where the foundry supports them.

DFT normally incorporates scan, BIST and test points so that post-manufacturing ATPG can detect stuck-at, transition and other defect classes. [23]

Given the 36 MiB cache hierarchy, **MBIST is mandatory in practice**. Attempting to functionally test every SRAM cell exclusively through CPU software would produce excessive tester time and poor diagnostic resolution.

CMOS wafer fabrication, BEOL and packaging

Front-end-of-line

Once mask data and a qualified process recipe exist, wafer fabrication begins. The exact sequence is proprietary to the foundry, but the conceptual CMOS flow is:

Isolation and wells. The wafer receives well formation and isolation structures, typically using implantation, oxidation/deposition, lithography, etch and annealing. Modern CMOS uses sophisticated shallow-trench isolation and multiple well structures.

Gate formation. Gate dielectric and gate-electrode structures are deposited/patterned. At a 28 nm HKMG node this is no longer simply the textbook thick SiO_2/poly-Si gate of early CMOS.

Source/drain engineering. Photoresist masks expose selected regions for ion implantation. Implant energy and dose determine the dopant distribution, while thermal anneals electrically activate dopants and repair lattice damage.

Contacts and local interconnect. Silicides/contact structures connect transistor terminals into the metallisation system.

ASML's process overview describes deposition, photoresist coating, exposure, bake/develop, etch, metrology and ion implantation as repeatedly executed operations during chip construction. [24]

Lithography

For each lithographic layer:

```
film deposition
      ↓
photoresist spin coat
      ↓
soft bake
      ↓
reticle alignment + exposure
      ↓
post-exposure processing
      ↓
development
      ↓
```

```
etch / implant / deposition
      ↓
resist strip + clean
      ↓
inspection / metrology
```

At 180/65/28 nm, DUV lithography can perform the relevant patterning with node-specific resolution enhancement and, where necessary, multiple-patterning techniques. At advanced nodes, mask optimisation and computational lithography become increasingly important. ASML notes that lithography patterns may deliberately be distorted computationally to compensate for optical and process effects. [25]

This is part of why **the designer should submit foundry-rule-clean GDS/OASIS, not attempt to design the reticle corrections manually**.

Etch, implantation and anneal

Dry plasma etch is used where anisotropic profiles and small critical dimensions are required; wet etch remains useful for appropriate materials and cleaning. ASML distinguishes dry gas-phase and wet chemical etching and emphasises the need to control multilayer structures without damaging underlying films. [26]

Ion implantation introduces dopants with controlled dose and depth. Subsequent thermal treatment repairs implantation damage and activates dopants. [24]

CMP

After deposition and patterning, surface topography accumulates. Chemical-mechanical planarisation removes excess material and restores a sufficiently flat surface for later lithography. The practical importance of wafer/local topology rises as dimensions shrink; Siltronic specifically identifies surface nanotopology and local flatness as important for lithographic and dielectric-planarisation performance at advanced dimensions. [27]

Back-end-of-line

The finished transistors are useless until connected. The BEOL stack repeatedly creates dielectric layers, vias and patterned metal routing.

For SilicaFlux, lower metal layers will carry dense local cell and SRAM routing; upper thicker metals should carry:

- clock trunks;
- long NoC/coherency links;
- high-current VDD/VSS mesh;
- memory-interface routing;
- package-bump connections.

After routing, dummy metal fill is inserted to satisfy density/CMP rules. Parasitic extraction then determines real resistance and capacitance, and STA/IR/EM analysis is repeated against the extracted layout.

DFM

Passing minimum DRC is necessary but not sufficient for high-yield manufacturing. DFM should include:

- lithographic hotspot analysis;
- recommended rather than merely minimum spacing where affordable;
- redundant vias on high-current and critical nets;
- metal-density optimisation;
- via-density checks;
- antenna protection;
- critical-area reduction;
- SRAM edge/abutment rules;
- package-bump escape checks;
- manufacturing pattern matching if supplied by the foundry.

Synopsys notes that design-rule count and complexity increase significantly as nodes shrink. [28] That is another argument against moving SilicaFlux directly to 7 nm before the architecture has been proven in silicon.

Wafer probe

After fabrication, probe cards contact each die while still on the wafer. Production test should include:

- scan ATPG;
- MBIST;
- PLL/clock tests;
- I/O continuity;
- leakage/current screening;
- basic functional boot;
- speed binning where required.

Advantest describes wafer-level electrical test as the stage where probe equipment and ATE verify die characteristics before dicing. [29]

Bad dies are mapped and excluded from packaging.

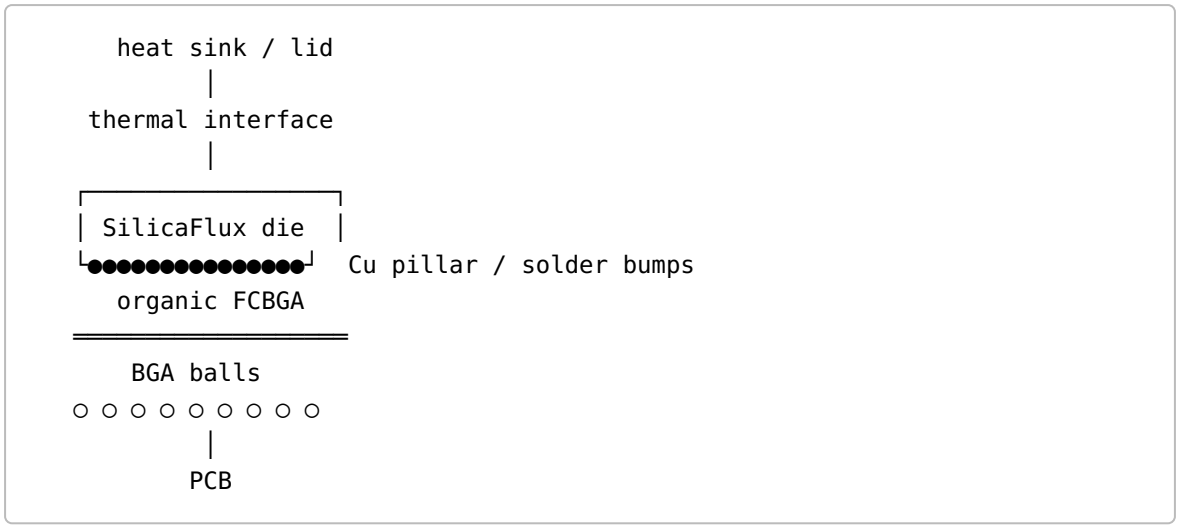
Packaging recommendation

For a reduced 180/65 nm prototype, a wire-bond QFP/BGA can minimise development cost.

For the full 28 nm SilicaFlux configuration, **flip-chip BGA is preferable** because the design needs substantial power delivery, memory bandwidth and many signals. Flip-chip allows power and ground connections across the die surface rather than only at the edge and shortens electrical interconnect. Amkor identifies lower signal and power inductance, higher I/O density and improved core power delivery as key flip-chip advantages. [30]

An FCBGA substrate can provide multiple high-density routing layers and a large bump array; Amkor's current FCBGA portfolio supports multi-layer substrates and fine-pitch Cu-pillar/flip-chip configurations intended for high-performance devices. [31]

A practical package stack would be:



Post-package ATE then repeats scan/memory tests and executes functional, speed, power and thermal screens. Advantest distinguishes wafer test, die-level test, final package test and system-level testing as separate test insertions. [32]

Production economics, schedule and yield

Prototype and NRE costs

Public MPW prices are useful because they expose the order-of-magnitude difference among nodes. EUROPRACTICE states that MPW combines designs from multiple customers and can cut prototype fabrication cost to roughly 5–10% of a dedicated prototyping run. [33] Its 2026 schedule lists, amongst other offerings:

Process	Public 2026 MPW indication
UMC 180 nm	€16,750 standard 5 × 5 mm block
UMC 65 nm	€43,750 standard 4 × 4 mm block
GF 28 nm SLPe	€12,430/mm² under published terms
GF 22 nm FDSOI	€17,820/mm² under published terms
GF 12 nm LP+	€27,500/mm² under published terms

[34]

These prices exclude much of the total product cost: engineering salaries, verification, EDA, IP licensing, SRAM/PLL/PHY licensing, packaging, probe cards, ATE programme development, boards and qualification.

A defensible **project-planning** budget—not a foundry quotation—would therefore be approximately:

SilicaFlux implementation	Engineering/NRE planning range	First-silicon fabrication/package/test
Reduced 180 nm demonstrator	£0.5–2m	£50k–250k
Reduced/fullish 65 nm	£1.5–5m	£150k–750k
Full 28 nm product	**£4–15m**	**£0.5–3m**
7 nm commercial derivative	£20–80m+	several £m upward

These are my budgeting estimates, not published foundry tariffs. They assume an existing engineering organisation and vary enormously with commercial EDA/IP arrangements. The public MPW escalation and the enormous capital required to manufacture advanced silicon provide the underlying scale evidence. [35]

Why building a fab is not part of the recommendation

A complete quartz-to-ASIC-owned factory would require entirely separate businesses:

- quartz qualification/mining;
- silicon smelting;
- chlorosilane chemical plant;
- polysilicon deposition;
- crystal-pulling and wafer finishing;
- wafer fab;
- mask shop or mask procurement;
- packaging;
- semiconductor test.

Even an industrial polysilicon cleaning expansion is capital intensive: WACKER reported an investment exceeding €300 million for a new semiconductor-polysilicon cleaning line. [36] A 22/28 nm foundry facility can cost billions; TSMC's Kumamoto 22/28 nm fab was announced at about US$7 billion. [37]

The correct economic model is therefore **buy material, buy foundry capacity and buy OSAT services**.

Yield model

Before physical implementation, exact SilicaFlux yield cannot be predicted. Foundry defect-density data, SRAM yield models and the actual die area are missing.

A useful first-order model is:

$$Y \approx e^{-D_0 A}$$

where D_0 is the assumed random defect density in defects/cm² and A is die area in cm². This model intentionally omits clustering, systematic yield loss, redundancy and parametric failures.

An illustrative—not predictive—comparison is:

Node	Illustrative SilicaFlux die area	Assumed D_0	Poisson random-defect yield
180 nm	400 mm²	0.10/cm²	~67%
65 nm	180 mm²	0.12/cm²	~81%
28 nm	**80 mm²**	**0.15/cm²**	**~89%**
7 nm	35 mm²	0.25/cm²	~92%

Those numbers are merely sensitivity analysis. Real foundry yield data are confidential and node maturity matters materially. TSMC publicly says N7/N7+ reached high-volume maturity and that N7+ achieved yields comparable with the preceding N7 process, but it does not disclose customer-specific die yield in the cited material. [38]

The more important conclusion is structural: **reducing die area improves the number of dies per wafer and reduces defect exposure**, but there is a point where extra process cost outweighs area savings. For SilicaFlux, 28 nm is likely near that economic balance.

Step-by-step production programme

A realistic production sequence for the current codebase is:

Phase	Indicative programme duration	Principal exit criterion
Architecture freeze	1–2 months	Fixed feature/cache configuration
RTL clean-up + verification	4–8 months	Regression, coverage, formal, CDC/RDC closed
ASIC synthesis exploration	1–2 months overlapping	Frequency/area/power target demonstrated
Foundry/IP selection	1–2 months overlapping	PDK, SRAM, PLL, I/O, package selected
DFT architecture	1–2 months	Scan/MBIST strategy frozen
Floorplan + power plan	1–2 months	Macro placement and IR feasibility
Placement/CTS/route	2–4 months	Routable design, clocks closed
Sign-off	1–2 months	STA/DRC/LVS/IR/EM/DFM clean
Tapeout/mask generation	weeks	Accepted GDS/OASIS
Wafer fabrication	roughly 3–4 months planning allowance	Completed wafers
Probe + package assembly	2–6 weeks	Packaged engineering samples
Bring-up + characterisation	1–3 months	Functional, PVT and reliability data

Phase	Indicative programme duration	Principal exit criterion
Production qualification	2–6+ months	Yield/test/qualification targets met

The wafer-fabrication allowance is consistent with the fact that modern chips require hundreds to thousands of repeated processing steps over a manufacturing cycle measured in months; ASML describes the manufacturing chain as involving hundreds of process steps and potentially several months. [24] MPW submission dates are discrete rather than continuous, so missing a shuttle can add additional calendar delay. Imec's 2026 service publishes specific final-GDS deadlines and requires advance registration for foundry runs. [39]

For **this codebase**, the dominant uncertainty is not wafer fabrication—it is getting to a trustworthy tapeout database. The earlier code assessment found assertions but relatively simple testbenches and no full UVM infrastructure, which makes verification closure the highest design-risk workstream. filecite turn0file0 L57-L91

Principal risks and mitigations

Risk	Probability before hardening	Consequence	Mitigation
RTL functional bug	High	Silicon respin	Reference models, constrained-random/UVM, formal
CDC/RDC defect	Medium-high	Intermittent silicon failures	Dedicated CDC/RDC sign-off
Cache coherency bug	Medium-high	Data corruption	Formal coherency properties + stress verification
SRAM test escape	High without MBIST	Yield/reliability loss	MBIST, redundancy, ECC
Timing failure	Medium	Frequency loss or respin	MMMC STA, post-route extraction
Tensor congestion	Medium	Unroutable/high power	Regular floorplan, datapath-aware placement
IR drop	Medium-high	Timing faults	Power mesh, bump planning, dynamic IR analysis
Electromigration	Medium	Lifetime failure	Current-density sign-off + redundant vias
PLL/noise coupling	Medium	Jitter/clock faults	Physical isolation and clean supplies
ESD failure	Medium	Low assembly/ customer yield	Qualified I/O/ESD cells and package-aware checking
DRC/LVS error	Low after proper sign-off	Mask failure	Foundry-certified sign-off decks

Risk	Probability before hardening	Consequence	Mitigation
Package power limits	Medium	Thermal throttling	FCBGA, thermal model, lid/ heat sink
Advanced-node NRE	High at 7 nm	Commercial failure	Start at mature 28 nm

Final recommendation

The optimal SilicaFlux production chain is therefore:

```
high-purity quartz
      ↓
carbothermic smelting
      ↓
~99% metallurgical silicon
      ↓
chlorosilane synthesis + fractional distillation
      ↓
Siemens deposition
      ↓
electronic-grade polysilicon
      ↓
CZ single-crystal growth
      ↓
300 mm polished prime wafer
      ↓
28 nm CMOS FEOL
      ↓
contacts + BEOL metal stack
      ↓
wafer probe
      ↓
dicing
      ↓
flip-chip FCBGA assembly
      ↓
final ATE
      ↓
system-level qualification
      ↓
production SilicaFlux ASIC
```

For the **first actual chip**, however, I would not tape out the current default configuration unchanged. I would first create two configurations:

SilicaFlux-P, the product target: eight cores, vector/tensor acceleration, full coherency, but resize the LLC after workload profiling—likely beginning around 8–16 MiB rather than committing immediately to 32 MiB.

SilicaFlux-V, the validation ASIC: one or two cores, one tensor/vector tile, a small L2/LLC, all clock/reset/DFT structures and representative SRAM macros. This die should be run on a low-cost MPW shuttle to prove the RTL-to-package flow, the clock/reset architecture, SRAM/MBIST implementation, I/O and software stack.

Only after SilicaFlux-V closes silicon bring-up should the full product move to a dedicated or larger MPW 28 nm implementation. This sequence sacrifices little architectural knowledge but substantially reduces the probability of paying for a large, cache-heavy first-silicon respin. The project's existing assessment already identifies verification maturity as its major gap, making this staged approach particularly important for this RTL rather than merely a generic ASIC recommendation. filecite turn0file0 L147-L176

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